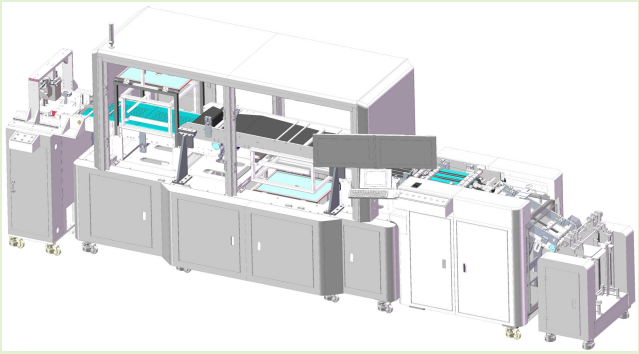
A Novel Hybrid YOLO Approach for Precise Paper Defect Detection With a Dual-Layer Template and an Attention Mechanism

Xuanfeng Li, Haibo Yan, Kai Cui, Zhiwu Li, *Fellow, IEEE*, Ruibin Liu, Guibin Lu, Kai Chin Hsieh, Xiaoshi Liu, and Chitin Hon, *Senior Member, IEEE*

***Abstract*—In a printing production process, paper often exhibits various surface defects due to objective factors. These defects manifest in different forms and are typically small and densely distributed, posing a challenge for existing object detection methods. To tackle this issue, this research reports a paper product appearance inspection system that combines a dual-layer template and an attention mechanism. The system effectively addresses the problem of surface defect detection by matching template images with the images to be inspected. This approach ensures robust detection performance and reduces false negatives (FNs) through the utilization of the improved You Only Look Once v7 (YOLOv7) algorithm. The enhanced algorithm incorporates diverse attention mechanisms that effectively utilize features, thereby enhancing the model's capability for defect detection. The system's development includes the creation of a visual interface that provides a customized defect detection solution for the printing authority. Through comprehensive ablation experiments, the algorithm demonstrates significant improvements in both detection accuracy and speed. This system enables automatic inspection of the appearance of single-page paper products and automates the removal of products with surface defects. Consequently, it achieves a high level of automation in the inspection and handling processes for single-page paper products.**



***Index Terms*— Dual-layer template matching, efficient channel attention (ECA) module, heat map, image processing, You Only Look Once (YOLO).**

NOMENCLATURE

MDE	Memetic differential evolution.
DE	Differential evolution.
EMDE	Enhanced MDE.
YOLOv7	You Only Look Once v7.
CNNs	Convolutional neural networks.
FMCC	Feature map cross correlation.
P	Precision.
SiLU	Sigmoid linear unit.
FReLU	Funnel rectified linear unit.
ReLU	Rectified linear unit.
PReLU	Parametric ReLU.
ROI	Region of interest.
SE	Squeeze-and-excitation module.
CBAM	Convolutional block attention module.
CA	Channel attention module.
ECA	Efficient channel attention modules.
CMOS	Complementary metal–oxide–semiconductor.
CCD	Compared with charge-coupled device.
FPS	Frames per second.
FPR	False positive rate.
FNR	False negative rate.
mAP	Mean average precision.

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Xuanfeng Li, Haibo Yan, Zhiwu Li, Ruibin Liu, Guibin Lu, Xiaoshi Liu, and Chitin Hon are with the Institute of Systems Engineering, Macau University of Science and Technology, Macau 999078, China (e-mail: 2220016449@student.must.edu.mo; 1909853smi30001@student.must.edu.mo; zwli@must.edu.mo; 2109853gmi30007@student.must.edu.mo; gblu@must.edu.mo; 1210017248@student.must.edu.mo; cthon@must.edu.mo).

Kai Cui is with the School of Computer Science and Engineering, Macau University of Science and Technology, Macau 999078, China (e-mail: 2109853nia30001@student.must.edu.mo).

Kai Chin Hsieh is with the Department of Mechanical Engineering, University College London, WC1E 7JE London, U.K. (e-mail: zcemhsi@ucl.ac.uk).

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TP	True positive.
FP	False positive.
TN	True negative.
FN	False negative.
R	Recall.
CAM	Class activation mapping.
Grad-CAM	Gradient-weighted CAM.

I. Introduction

PRINTED products, encompassing newspapers, books, magazines, reports, documents, and packaging, hold significant importance in both individuals' daily lives and professional endeavors. However, the production process of paper printing is susceptible to the emergence of surface imperfections on these materials, which can arise from external environmental pollution, equipment contamination, and mishandling by printing personnel [1]. These flaws have the potential to compromise the overall quality, user experience, and visual allure of the paper, consequently diminishing its economic worth [2]. Therefore, the detection of paper defects assumes a vital role in the papermaking process.

In the paper manufacturing industry, both physical quality indicators and appearance indicators hold equal significance for paper products. However, in practical production processes, several common surface defects arise, including black spots, bright spots, holes, wrinkles, cracks, and blocks, as depicted in Fig. 1.

These defects have a notable impact on the brightness, smoothness, and uniformity of the paper [3]. Black spots and bright spots refer to visible blemishes and streaks on the paper surface, exhibiting distinct color variations compared to the surrounding area. The origins of these spots are diverse and complex, stemming from factors such as raw materials, improper cooking, coarse screening, fine screening processes, or operational and equipment issues. Holes are a prevalent defect in paper and indicate the areas where fibers have not fully penetrated the paper sheet. They are typically classified as small holes, slits, or perforations, collectively known as holes. Wrinkles, characterized by uneven tension during the paper production process, manifest as dark lines or areas in paper images. Detecting wrinkles has proven challenging due to their low contrast. Wet wrinkles and dry wrinkles are the primary types, distinguished by their inherent characteristics [4]. Wrinkles exhibit a linear pattern, allowing for the utilization of line detection methods in their detection. Cracks, typically located in the middle or edges of the paper, are another prevalent defect. Longitudinal cracks are more commonly observed in the middle section, while transverse cracks tend to occur at the edges.

In the realm of printing industry, the detection of defects plays a pivotal role in upholding the quality of printed materials [5]. Traditional manual inspection has long been employed for surface defect detection. However, within modern industrial systems, manual inspection is plagued by significant limitations. It heavily relies on visual scrutiny conducted by inspection workers stationed at their respective workstations, who manually eliminate products exhibiting surface defects upon identification. This removal process may involve the use of hands or tweezers to extract defective items from the conveyor belt. Nonetheless, the manual inspection method encounters various constraints. First, the human eye possesses limited spatial and temporal resolution, rendering the detection of minute-scale defects and inspections in high-speed production environments challenging [6]. Second, manual inspection proves inefficient and unsuitable for large-scale industrial production. Third, the inherent variability among individuals performing manual inspections may result in inconsistent detection standards, thereby impacting the consistency of product specifications. Consequently, there exists an urgent imperative to explore paper inspection methods tailored to the requirements of small- and medium-sized printing enterprises.

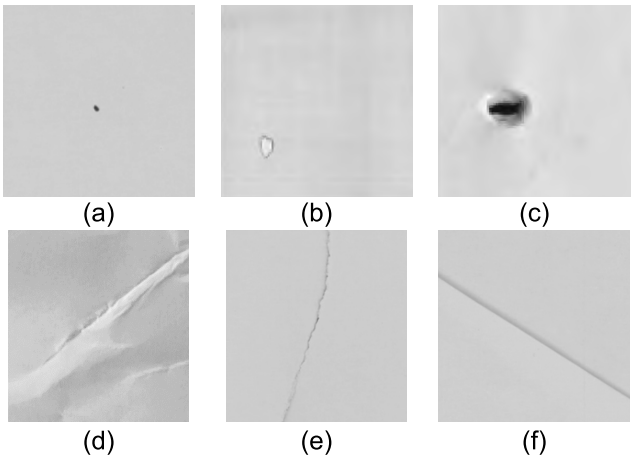


Fig. 1. Various degrees of surface defects. (a) Black spot. (b) Bright spot. (c) Hole. (d) Wrinkle. (e) Crack. (f) Block.

With the rapid progress of machine vision and industrial automation, the concept of Industry 4.0 has emerged as a prominent focal point [7]. Numerous conventional manual operations are gradually being supplanted by machines. Undoubtedly, this trend represents an inevitable path in technological advancement and development, as the ultimate objective of industrial progress is to enhance the quality of human life [8]. An MDE algorithm was proposed for the design of digital filters, aiming at detecting defects in paper production processes. This algorithm synergistically combines the exploratory capabilities of DE with the developmental characteristics of two local search techniques. The results demonstrate the efficacy of the DE framework in addressing the investigated problem category while averting stagnation in the DE exploration mechanism [9]. Tirronen et al. [10] introduced an EMDE for digital filter design. The algorithm incorporates the utilization of two Gabor filters and the execution of EMDE. EMDE represents a novel adaptive evolutionary algorithm that amalgamates the robust exploration attributes of DE with the developmental features of three distinct local search algorithms, each adopting unique pivot rules and neighborhood generation functions. Numerical findings substantiate the superior performance of EMDE. A rotating terahertz imaging system was proposed to tackle the potential occurrence of minor defects on the rear surface of products during the initial forming process. This system can be seamlessly integrated into the manufacturing process to generate high-resolution

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images of the casing, thereby facilitating discussions regarding anomaly detection employing this methodology [11].

Recently, there has been a rapid development of diverse visualization-based intelligent techniques for the detection and classification of surface defects on paper. However, the majority of advanced technologies are customized for large-scale paper manufacturing companies, equipped with costly machinery that suits their production lines. At present, in China, there is a focus on promoting low-carbon and environmentally friendly practices in the paper industry. As a result, the majority of paper mills in the country are privately owned enterprises of small to medium scale. These enterprises have specific practical requirements for paper surface defect detection, which include high recognition rates, simple methodologies, and cost-effective equipment solutions [12].

The primary objective of this article is to develop a versatile printing defect detection system that encompasses mature software and hardware technologies. The system is designed to align with the practical production requirements of the printing industry, considering both its universality and robustness. By simplifying the setup process, the system allows even nonprofessionals to train and utilize defect detection models for various types of printed products, as guided by the provided instructions. Widespread adoption of this system within the industry holds significant practical value. The contributions of this article can be summarized as follows.

1) We present a novel paper product appearance inspection system that integrates a dual-layer template and an attention mechanism. The system matches template images with the images to be inspected and utilizes an improved YOLOv7 algorithm, incorporating different attention mechanisms to fully leverage features and enhance the model's detection capabilities.

2) The system's development includes the creation of a visual interface that provides a customized defect detection solution for the printing bureaus. By automating the inspection process, it enables the automatic detection and removal of defective single-page paper products, eliminating the need for manual inspection and removal. This not only reduces labor costs but also improves overall work efficiency for printing bureaus. In addition, it facilitates automation in the inspection and handling of paper product appearances, ensuring consistency in the final output.

3) For longer paper products, the system employs a controller and paper detection sensor to precisely determine the position of the paper before capturing the image. This ensures complete image acquisition for longer paper products and overcomes limitations associated with the field of view of area scan cameras.

4) This study aims to address the issue of unwanted artifacts, specifically conveyor belts, in the background of camera-captured images. To overcome this challenge, this article proposes a Python-based image preprocessing technique. The technique involves image reading, size adjustment, edge detection, contour finding, and image transformation.

This article is structured as follows. In Section II, the extensive research and application of deep learning algorithms in the domain of surface defect detection are briefly discussed. Subsequently, in Section III, the activation function, dual-layer templates, and attention mechanism are covered. Section IV introduces the hardware system employed in this study, along with an enhanced YOLO network model. Section V details the preprocessing steps for the experimental dataset and outlines the evaluation metrics. In Section VI, this article presents comprehensive ablation experimental results, with a focus on the development of visual interface software and the application of heat maps for in-depth discussion and verification. Section VII is dedicated to a broader discussion. Finally, Section VIII summarizes the conclusions and outlines potential future directions.

II. Related Work

Machine vision-based industrial defect detection pertains to the utilization of visual devices in the industrial domain to analyze, detect, identify, and classify surface defects in industrial products, thereby supplanting labor-intensive and error-prone manual visual inspection. A method based on multiscale wavelet transform and Gaussian mixture model was presented for the automated detection of fabric defects [13]. The sample images underwent wavelet decomposition and subsequent denoising via wavelet thresholding. Subsequently, the resulting images were segmented using a Gaussian mixture model implemented through the expectation–maximization algorithm. Experimental findings showcased the algorithm's capacity to accurately identify defect locations and segment the defects. Defect detection algorithms relying on feature comparison, characterized by their intricate image preprocessing and limited robustness, encounter challenges in terms of computational load and accurate detection of defect size and shape [14], [15]. Such algorithms are generally suitable for regular defects and simple scenes but prove less applicable in situations involving indistinct features, diverse shapes, and complex scenes.

With the advancement and maturation of CNNs, deep learning algorithms have been extensively studied and applied in the domain of surface defect detection [16]. CNNs possess the capability to update parameters through sample training, eliminating the need for the manual design of complex algorithms and offering high levels of robustness and accuracy. Deep neural network-based defect detection methods exhibit advantages over manual inspection methods, including precise defect recognition and immunity to fatigue. Once a detection system is established, it can consistently and automatically detect defects based on a predefined standard, without overlooking any [17], [18]. Another approach aimed at detecting flaws in printed materials involves the utilization of scale-adaptive template matching and image alignment. Initially, this methodology employs a CNN to extract profound feature vectors from both the template and the low-resolution target image. Subsequently, the likeness between the template and the target image is assessed by employing FMCC as a metric for matching. Ultimately, through the utilization of the image alignment module, the detection of print defects is executed, resulting in an impressive accuracy rate of 93.62% [19].

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TABLE I
COMPARATIVE ANALYSIS OF TWO MAINSTREAM MODELS

Network model	Precision(%)	mAP(%)	FPS
Faster-RCNN	**84.2**	**89.1**	25
YOLOv7	83.2	87.3	**82**

A deep learning-based Faster R-CNN method is introduced [20], which leverages deep features extracted from defect targets to effectively address the challenge of detecting internal joint defects, a task with which traditional algorithms struggle. This method also demonstrates applicability in various lighting conditions, reducing the need for preprocessing steps, and improving efficiency. The researchers applied the Faster R-CNN method to adapt it to their fabric defect dataset, aiming to address the issues of time-consuming and inefficient traditional detection methods. The improved Faster R-CNN model reduces the number of anchor points, utilizes feature maps in the ROI pooling layer to extract proposal features, and performs classification using subsequent fully connected and softmax networks. Experimental findings highlight the capability and effectiveness of the improved Faster R-CNN model in fabric defect detection [21]. However, deep neural network-based defect detection methods encounter limitations such as weak generalization ability, longer computational time, and the requirement for a substantial amount of training data encompassing various types of defects [22].

Due to the necessity of pregenerating candidate regions, Fast R-CNN evinces a relatively languid pace. In striking contradistinction, YOLO flaunts rapid detection prowess, making it eminently suitable for real-time object detection deployments. Nonetheless, its precision (P) bears certain constraints, as delineated in Table I. Consequently, we have embarked upon the task of enhancing the YOLO algorithm, pursuing not only heightened celerity but also the attainment of commendable precision in its outcomes.

III. PROPOSED METHOD

A. Activation Function

This article replaces the SiLU activation function in the Conv layer with Funnel ReLU (FReLU) to introduce spatial conditions into the nonlinear activation of each pixel [23]. This change enables the modeling of pixel-level spatial information through $T(x)$ spatial conditions, allowing for the capture of complex visual layouts using regular convolutions. By comparing several common activation functions such as ReLU, PReLU, Swish, and FReLU, it is observed that FReLU is more efficient and accurate in tasks such as classification, detection, and segmentation. FReLU also demonstrates good transferability, as depicted in Fig. 2. The specific calculation formula is as follows:

$$\mathrm{ReLU} = \max(0, x) \tag{1}$$

$$\mathrm{PReLU} = \max(0, px) \tag{2}$$

$$\mathrm{SiLU} = x * \mathrm{sigmoid}(x) \tag{3}$$

$$\mathrm{Swish} = x * \mathrm{sigmoid}(\beta x) \tag{4}$$

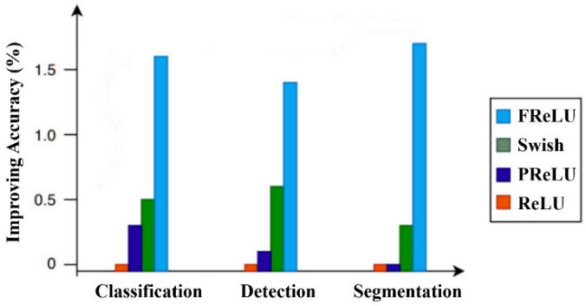


Fig. 2. Activation function performance comparison.

$$\mathrm{FReLU} = \max\left(x_{c,i,j}, T\left(x_{c,i,j}\right)\right) \tag{5}$$

$$T\left(x_{c,i,j}\right) = x^{w}_{c,i,j} \cdot p^{w}_{c} \tag{6}$$

where PReLU signifies ReLU with parameters and Swish is represented as SiLU with parameters. When $\beta = 1$, Swish is equivalent to SiLU. $T(x)$ represents the 2-D spatial capability, and $x^{w}_{c,i,j}$ represents the coordinates of the cth channel in the 2-D space of i and $j \cdot p^{w}_{c}$ denotes the coefficient on these coordinates.

B. Double-Layer Template

Template matching is a technique for finding the most similar part of the source image (S) with respect to a template image (T). Templates are small images that are already available, while template matching is the process of searching for objects in large images, and our objective is to detect the best matching regions. The target in the image has the same size, orientation, and image characteristics as the template. The coordinate position of the target in the image can be determined by a specific algorithm. Through template matching, we can utilize a "template" containing the object of interest to detect the object within the input image. The process of template matching involves sequentially comparing the source image with templates stored in a library, storing the comparison results in a matrix, and estimating the best match position of the template within the dataset using a scoring algorithm. In essence, this method employs a 2-D sliding window that traverses the image to be examined, moving pixel by pixel from left to right and top to bottom. The algorithm calculates the degree of pixel matching between the template and the overlapping subimage. A higher degree of matching indicates greater similarity and the brightest position indicates the optimal match location.

1) Template Matching Based on Gray Value: Template matching based on gray value is the most classical algorithm for template matching. Its principle is as follows. First, an ROI is selected as the template image and a grayscale-based template is generated. Next, a coarse matching is performed by comparing the detection image with the template image. By selecting an arbitrary point in both the detection and template images and employing the method of point-wise comparison, the similarity of their grayscale values is calculated, thereby obtaining approximate correlation points. Subsequently, we proceed with precise matching by taking the rough corresponding points as the central points and utilizing

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the least squares method to find the optimal matching points. This study employs either the absolute error and sum of absolute differences [24] or the sum of squared differences to calculate the distance between the pixel grayscale differences of the template and detection images for localization. Assuming that $S(x, y)$ is the search image of size $m \times n$ and $T(x, y)$ is the template image of size $M \times N$, the objective is to find the region in the search image S that matches the template T. Within the search image S, starting from the top-left corner (i, j), a subimage of size $M \times N$ is selected, and its similarity with the template is computed. By traversing the entire search image and considering all possible subimages, the subimage that is most similar to the template image is identified as the final matching result. The specific calculation formula is as follows:

$$D(i, j) = \sum_{S=1}^{M} \sum_{t=1}^{N} |S(i + s - 1, j + t - 1) - T(s, t)| \tag{7}$$

$$D(i, j) = \sum_{S=1}^{M} \sum_{t=1}^{M} |S(i + s - 1, j + t - 1) - T(s, t)|^2 \tag{8}$$

where

$$1 \leqslant i \leqslant m - M + 1, \quad 1 \leqslant j \leqslant n - N + 1. \tag{9}$$

Since this method uses all the gray values of the template image for matching, the gray value will change significantly when the lighting conditions change. Consequently, it is unable to adapt to lighting variations and is generally unsuitable for matching multichannel images. Moreover, it is primarily applicable to straightforward image matching scenarios. In order to solve the above problems, we have installed three bar-shaped light sources in each set of appearance visual detection devices to ensure the stability of lighting intensity.

2) Template Matching Based on Gradient of Edge Orientation: The purpose of edge detection is to locate positions of significant brightness variations within digital images. The template matching algorithm based on edge direction gradients utilizes the gradient correlation of object edges as the matching criterion. Its principle involves extracting edge features from the ROI [25] and creating a template by combining grayscale information. A multilevel image pyramid model is generated based on the size of the template and the desired level of clarity. Starting from the top level of the pyramid, the algorithm performs a top-down search for the template image, progressively moving to lower levels of the pyramid, searching for matching results at each level, until reaching the bottom level or obtaining a definitive match.

By using the Canny function and the FindContours function, we can detect the edges of the regions of interest in the image and obtain their coordinate values. Then, the Sobel function is employed to calculate the gradient values of the edge points in the x- and y-directions. Using the CartToPolar function, we can determine the gradient magnitude and gradient angle for each edge point. During the traversal of the edge points, we save the x and y gradient values for each point and perform normalization cross correlation on the gradient magnitudes. In addition, we convert the coordinates of the edge points to relative coordinates with respect to the center point.

This method leverages edge features to locate objects and exhibits robustness when dealing with factors such as lighting variations, image grayscale changes, local edge loss, cluttered scenes, noise, defocus, and slight deformations. It also supports simultaneous search for multiple templates, thereby enhancing the accuracy. However, in cases where the target image undergoes significant rotation or scaling, it may have an impact on the search results. Hence, this method is not suitable for handling scenarios involving substantial rotation or scaling.

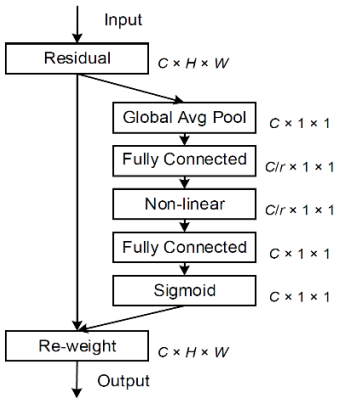


Fig. 3. SE module.

C. Attention Mechanism

The attention mechanism, a technique employed in data processing, emerges as one of the fundamental faculties of human perception and decision-making. In the realm of cognitive science, due to the constraints of information processing, individuals selectively attend to a segment of data while disregarding other visible information, thereby effectively harnessing limited resources for visual information processing. By allocating weights to different segments of the data, the attention mechanism autonomously selects and concentrates on pivotal information within the input data. In the domain of machine learning, the attention mechanism finds extensive application in augmenting the performance of deep neural networks. This article will introduce several sophisticated attention mechanism modules. These modules possess the capability to extract the indispensable feature information required for the given task from input images, with a specific emphasis on their applications in the realm of computer vision.

1) SE Module: The SE module [26], depicted in Fig. 3, functions as a universal attention mechanism module. It utilizes a fully connected network to acquire feature weights, thereby enabling the network to concentrate on the most pivotal features and amplify the weights of impactful feature channels. The operational procedure unfolds as follows: the features of each channel undergo global average pooling, two fully connected layers are employed to derive weights for the channels, and the ultimate feature representation is obtained by multiplying the features with the respective weights.

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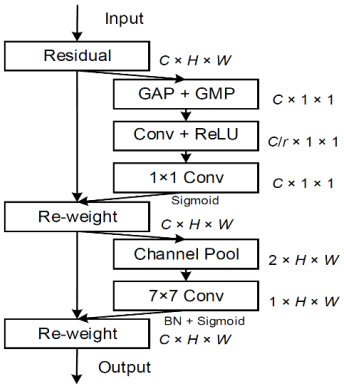


Fig. 4. CBAM module.

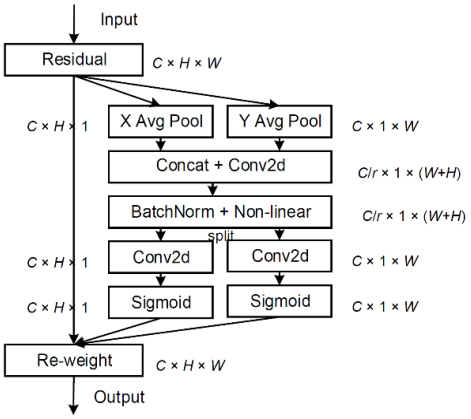


Fig. 5. CA module.

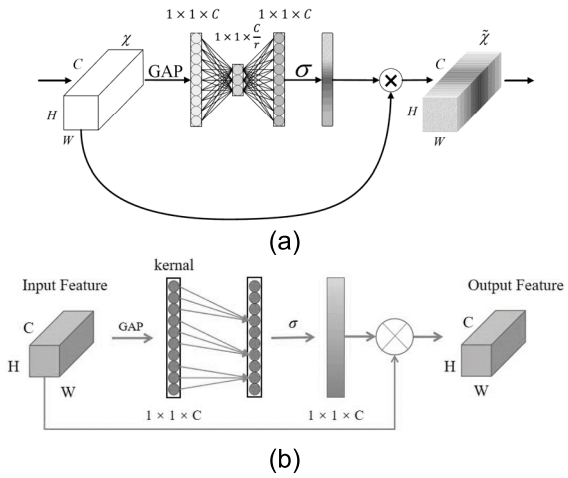


Fig. 6. Comparison of (a) SE module and (b) ECA module.

2) CBAM Module: The CBAM module amalgamates CA and spatial attention [27], as illustrated in Fig. 4. By deducing attention weights and multiplying them with the original feature map, it dynamically adjusts the features. This approach facilitates more efficient feature acquisition by taking into account the significance of both channel and spatial positions. The CBAM module captures comprehensive contextual information at both global and local scales, thereby amplifying the model's perception of crucial features.

3) CA Module: The CA module is an advanced multi-head self-attention mechanism [28], as illustrated in Fig. 5. By employing self-attention, it enables the network to learn the dependencies between different regions in the image, thereby enhancing the model's perception of channel correlations and ultimately improving its performance.

4) ECA Modules: The ECA module is a local attention mechanism module [29], as depicted in Fig. 6. In comparison to the SE module, the ECA module has a lower computational cost while effectively mining feature information. It applies a local attention mechanism by learning the relationships between pixels at each position. SE attention is computed using two fully connected layers, while ECA attention is based on neighboring channels with $k = 5$. The parameter k is crucial in determining the range and convergence of local interactions, and its selection depends on the specific CNN architecture or the number of channels in different scenarios.

In the ECA module, a 1 ×1 convolution layer is directly utilized after the global average pooling layer, eliminating the need for fully connected layers. This design not only avoids dimension reduction but also effectively captures inter-channel interactions. In addition, the ECA module has fewer parameters while yielding favorable results. It enhances the model's understanding and evaluation of channel features, thus improving overall performance [30]. The specific calculation formula is as follows:

$$k = \varphi(C) = \left| \frac{\log_2(C)}{\gamma} + \frac{b}{\gamma} \right|_{\text{odd}} \tag{10}$$

where C represents the number of input feature channels and $|x|_{\text{odd}}$ refers to the nearest odd number to the value of x. The values of γ and b are set as 2 and 1, respectively.

5) Model Selection and Collection: The application of attention mechanism techniques, such as SE, CBAM, CA, and ECA, can enhance the model's expressive power and generalization performance, allowing it to focus on important features and achieve better experimental results across various domains [31]. These attention mechanisms are suitable for different model architectures, but their selection and integration should consider the specific requirements and task demands of the model. For example, the SE module enhances the importance of channel features, the CBAM module captures global and local contextual information, the CA module focuses on dependencies between channels, and the ECA module efficiently models CA.

To evaluate the impact of the proposed attention mechanism improvements on paper defect detection and localization performance, we conducted ablation experiments to verify the detection performance of various attention mechanisms. In these experiments, each improvement algorithm was trained for 300 epochs, and the best results were chosen for comparison. The experimental results are presented in Table II, where the "√" symbol in Table II indicates the addition of corresponding modules within the original YOLOV7 network. The incorporation of the ECA module has yielded a notable enhancement in mAP when contrasted with other attention mechanisms.

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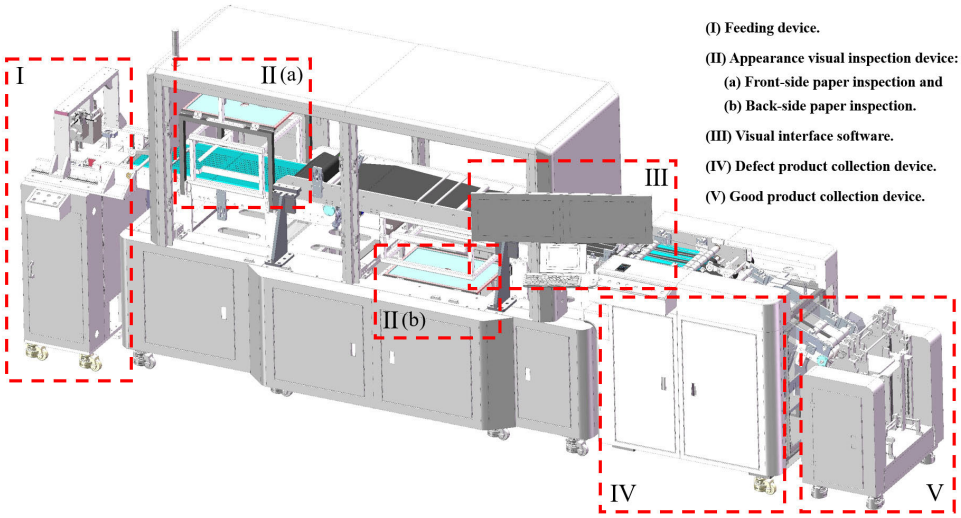


Fig. 7. Paper defect detection system.

TABLE II
RESULTS OF IMPROVEMENTS MADE TO ATTENTION MECHANISMS

Attention Mechanism				mAP(%)
SE	CBAM	ECA	CA	
×	×	×	×	87.3
√	×	×	×	83.6
×	√	×	×	88.9
×	×	√	×	**89.6**
×	×	×	√	86.9
√	√	×	×	85.6
√	×	√	×	86.7
√	×	×	√	84.3
×	√	√	×	87.6
×	√	×	√	86.5
×	×	√	√	85.8

When utilizing these attention mechanism techniques, it is crucial to consider various aspects such as model selection and integration, training data and annotations, hyperparameter tuning, and model interpretation and explainability. In particular, it is important to note that the effectiveness of attention mechanisms is influenced by the quality and accuracy of the training data. Therefore, it is essential to ensure that the training dataset is sufficiently large, representative, and properly annotated to allow the model to learn accurate attention distributions and weights. In addition, some attention mechanism techniques may require hyperparameter adjustments to enhance model performance through cross validation and experimentation. Finally, although attention mechanisms aid in explaining the model's decision-making process, it is essential to pay attention to the accuracy and consistency of the explanations. Understanding the relationship between attention distributions and model predictions is crucial for evaluating the reliability and interpretability of the model.

IV. IMPLEMENTATION OF A SURFACE DEFECT SYSTEM FOR PAPER BASED ON CNNS

A. Design and Control Methods for the System's Workflow

To attain optimal and precise outcomes within a printing defect detection system, it becomes imperative to possess image acquisition of superior quality, swift and efficient image preprocessing, as well as defect recognition algorithms. These constituents depend on an adept software and hardware infrastructure for reinforcement. The deep learning-based printing defect detection method serves as the central module within the comprehensive machine vision defect detection system. The quality of its design directly affects the performance and stability of the detection system.

A novel system, combining dual-layer templates and attention mechanisms, is proposed in this research for the visual inspection of paper products, aiming to effectively address surface defects encountered in print production. This device ensures the uniform presentation of the paper sheets upon their departure from the manufacturing facility. Its constituents encompass a framework, a feeding device, a primary conveyor platform, a secondary conveyor platform, a tertiary conveyor platform, an appearance visual inspection device, a waste removal mechanism, and a collection unit, as depicted in Fig. 7. By employing this equipment, the automated scrutiny and manipulation of the visual characteristics of individual paper sheets can be efficiently accomplished, thereby augmenting both production efficiency and standards of quality. The introduction of this groundbreaking technology has ushered in notable enhancements and advancements to the paper manufacturing industry.

The process of detecting and controlling paper defects is carried out through the utilization of a controller, as depicted in Fig. 8. In the initial stage, the controller receives signals pertaining to detection from the paper detection sensor. Based on the timing of the received signals, the controller computes the precise position of the paper product. Upon the paper product's arrival at the designated position, the controller dispatches a trigger signal to the appearance visual inspection device, thereby commencing the capturing process. This process and control method effectively harmonize the operations of the diverse components, ensuring that the capturing action transpires at the exact moment, consequently heightening the precision and efficiency of the appearance inspection of the paper product.

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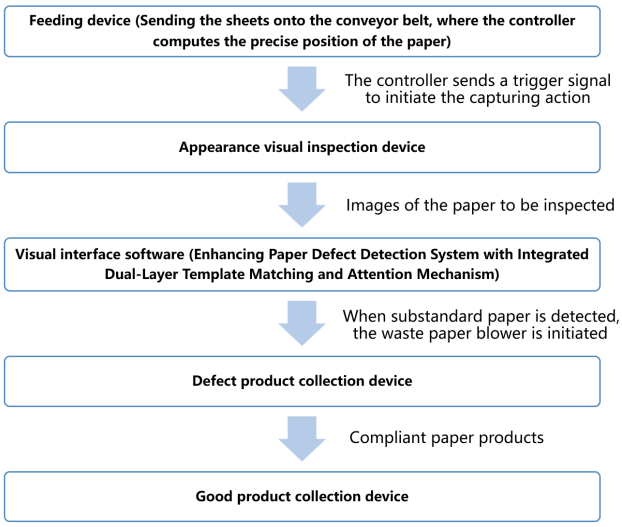


Fig. 8. Process and control procedure for inspecting and handling the appearance of a single-page paper product.

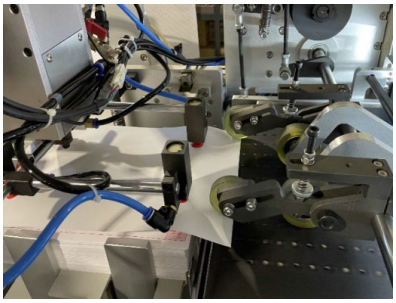
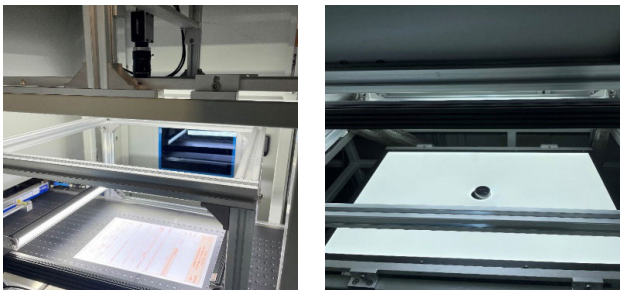


Fig. 9. Feeding device.

In the case of sample sheets that exhibit paper defects, following preprocessing and morphological operations, we extract their shape feature parameters. Subsequently, we employ a trained YOLOv7 neural network to establish a correlation between these parameters and the respective types of paper defects. Similarly, for the sheets undergoing inspection, we subject them to preprocessing and morphological operations and employ the trained network to discriminate the parameters, thereby achieving the identification of paper defect types. By acquiring the detection result signals from the appearance visual inspection device and scrutinizing the received signals, we can ascertain the qualification status of the paper product. When an unqualified paper product reaches the rejection position, we transmit a jet signal to activate the waste blowing fan, thereby initiating the flow of jet air. This process effectively facilitates the recognition and handling of paper defects, ultimately augmenting the quality and uniformity of paper products.

B. Hardware Components of the Detection System

1) Feeding Device: The operational platform of the feeding device is outfitted with adjustable components, a supportive framework, adsorption device components, a storage bin, a lifting adjustment apparatus, and paper positioning components, as depicted in Fig. 9. This equipment utilizes the lifting adjustment device to modify the height position of the storage bin, ensuring harmonization between the paper inside the storage bin and the adsorption device components through the paper positioning components. Consequently, the storage bin can cater to the feeding requisites of paper materials of varying sizes and quantities, facilitating the convenient adherence and conveyance of the paper by the adsorption device components onto the conveyor line [32]. This seamless feeding and transportation of paper materials effectively forestall issues such as paper abrasion, distortion, and adhesive overflow, thereby optimizing the caliber of the paper products.



(a) (b)

Fig. 10. Appearance visual inspection device. (a) Front-side paper inspection. (b) Back-side paper inspection.

The storage bin in the feeding device is used to store the paper to be tested. It possesses the capability to accommodate the diverse feeding requirements of paper materials in terms of sizes and quantities. The dimensions of the paper form a rectangular shape, with the length and width falling within the range of 200–400 mm. In the case of lengthier paper products, a collaborative effort between the controller and the paper detection sensor is employed. By precisely calculating the position of the paper product based on the timing of the received detection signals, it ensures that the longer paper products have completely reached the designated capturing position of the visual inspection device before the commencement of image capture. This design facilitates the acquisition of comprehensive image information for longer paper products and eliminates concerns pertaining to insufficient field of view when utilizing area scan cameras. Such a design guarantees the acquisition of accurate image data throughout the inspection process, ultimately enhancing the precision and dependability of the visual inspection conducted on the appearance of paper products.

2) Appearance Visual Inspection Device: The appearance visual inspection device comprises two sets of equipment, as illustrated in Fig. 10. Each set comprises an installation bracket, three strip light sources, and a camera. The purpose of the light sources is to provide uniform illumination to the designated area of the paper products undergoing inspection. Hence, fluctuations in external lighting conditions do not impede the accuracy of defect detection. The first set of cameras is specifically designated for evaluating the print quality of the paper, while the second set is utilized for detecting surface scratches or marks. This arrangement facilitates simultaneous assessment of both the print quality and surface conditions of the paper, effectively meeting the production requirements of the paper.

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TABLE III
SPECIFICATIONS OF THE USED CAMERA

Camera parameters	Details
Sensor	Gpixel GMAX0505
Pixel size	2.5 μm × 2.5 μm
Sensor resolution	5120 × 5120
Lens mount	C-mount
Dimensions	29mm × 44mm × 59mm

The two sets of inspection devices are positioned above the first conveying platform and beneath the second conveying platform. Their role is to inspect the front and back sides of the individual paper sheets. The conveying platforms employ belt conveyors, with the second conveyor belt being equipped with porous holes and negative pressure functionality. A negative pressure supply device ensures that the paper adheres to the underside of the second conveyor belt and slides in relation to it. This enables the second set of inspection devices to conduct inspection of the paper's backside in a bottom-to-top manner.

In order to achieve efficient and accurate print defect detection, high-quality image capture is essential. This article introduces the visual inspection device, also known as the CMOS visual detection device [33], to inspect the surface quality of the products. CMOS technology offers advantages such as high yield, high integration, low power consumption, and low cost. The specific model used in this context is listed in Table III. The Gpixel GMAX0505 camera, with its remarkable 5120 × 5120 pixel high resolution, is exceptionally well-suited for applications that demand high levels of detail, such as paper defect detection. Its small 2.5 × 2.5 μm pixel size enhances image quality, particularly in low-light conditions, aiding in capturing fine details and improving performance in low-contrast scenarios. Furthermore, the Gpixel GMAX0505 camera features a C-mount lens interface, making it compatible with a wide range of C-mount lenses, offering increased flexibility to cater to various application requirements, including options for focal length and lens type.

First, the CMOS image capture device converts a captured first, and the CMOS image capture device converts a captured object into image signals [34], which are then sent to a dedicated image processing system. The image processing system transforms the image signals into digital signals based on pixel distribution, brightness, color, and other information. By performing various operations on these signals, such as extracting area, quantity, position, length, and other features, the image processing system enables automatic identification. The system can discriminate the targets based on preset allowable ranges and other criteria, including dimensions, angles, quantities, pass/fail status, and presence/absence.

In contrast to CCD technology, CMOS cameras are known for their lower power consumption and superior integration. CMOS employs individual photodetectors with independent conversion controllers, allowing for fast readout speeds. CCD technology, on the other hand, utilizes sequential photodetection, resulting in slower speeds. High-speed cameras with frame rates above 500 frames/s predominantly utilize CMOS technology due to its faster readout capabilities.

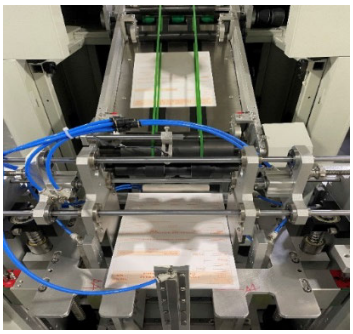


Fig. 11. Good product collection device and defect collection device.

Compared to other camera technologies, such as infrared cameras, high-speed cameras, and specialized cameras, CMOS cameras offer a variety of configuration options, including resolution, exposure time, and gain. This adaptability makes them suitable for diverse applications, allowing for adjustments as needed. In addition, CMOS cameras are typically more cost-effective, contributing to reduced overall expenses and enhancing their feasibility across multiple application domains.

3) Good Product Collection Device and Defect Collection Device: The collection device comprises two integral components: the good paper collection device and the defective paper collection device, as depicted in Fig. 11 [35]. The waste blowing device is positioned above the gap between the discharge end of the second conveying platform and the feeding end of the third conveying platform, while the defective paper collection device is situated below the said gap.

Upon detection of a defective paper by the visual inspection device, the detection result is transmitted to the controller. Subsequently, the controller activates the waste blowing device, directing a stream of air toward the gap, thereby causing the defective paper to be expelled into the gap. Following this, the defective paper is collected within the defective paper collection device. Conversely, when the visual inspection device determines that a paper meets the standards of good quality, it continues to be conveyed by the third conveying platform and ultimately reaches the good paper collection device. This device enables the automated collection of paper, eliminating the necessity for real-time manual collection of defective paper [36], [37]. As a result, it reduces the workload of operators, enhances efficiency, and minimizes the frequency of manual intervention in the collection process, thereby further improving the overall inspection efficiency.

C. Software Design of the Detection System

1) Software Architecture Diagram: The software architecture diagram of this system is illustrated in Fig. 12. Initially, it encompasses the collection of paper defect data and subsequent data annotation. Subsequently, an improved YOLOv7, combined with a dual-layer template algorithm, is employed to train the defect model. The trained model undergoes quantization to enhance recognition accuracy during practical application. Following this, the model is evaluated to examine

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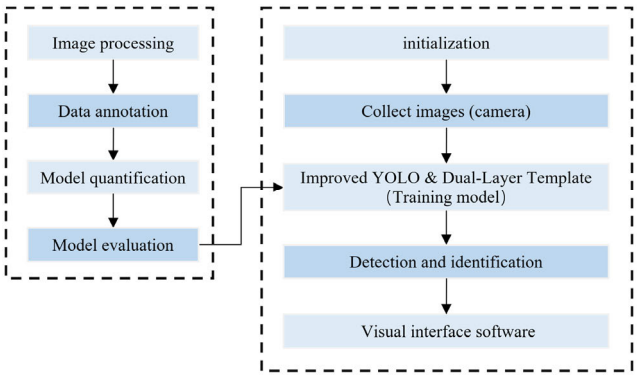


Fig. 12. Software architecture diagram.

whether there is any loss in accuracy after quantization and the extent of any such loss. If the evaluation confirms that the model's recognition accuracy remains within an acceptable range, the model is transformed and deployed onto embedded terminal devices.

2) Improved Network Structure Diagram: This study is centered around improving the performance of YOLOv7 [38]. To overcome challenges such as missed detections, FPs, and inadequate feature extraction for small object detection [39], we have introduced distinctive attention mechanisms in the head. Through extensive ablation experiments, we have determined that the ECA module, depicted in Fig. 13, is the optimal choice for integration into the head of the network structure. The ECA module boasts a modest computational burden, yet it efficiently excavates feature information, enhancing the model's comprehension and evaluation capabilities of channel features, thereby ameliorating the model's performance.

V. Experiment

A. Experimental Dataset

The paper dataset in this article was collected by two cameras at the Printing Bureau of Macao Special Administrative Region of the People's Republic of China. The labeling process involves three primary stages: image processing, annotation, and data conversion, as shown in Fig. 14. The collected printed image data underwent a series of steps, including screening, integration, and preprocessing to remove impurities such as conveyor belts. The processed dataset was then used to construct the training dataset, comprising labeled positive and negative samples. In order to increase the sample size and enhance the model's generalization capability, intentional defects were introduced into the normal samples during the training process. Through training the print defect detection model on this dataset, the defect judgment threshold was determined, enabling successful detection and classification of six distinct surface defects: black spots, bright spots, holes, wrinkles, cracks, and blocks. To annotate the paper defects, the LabelImg annotation software was utilized, resulting in the creation of a corresponding paper defect dataset [40]. This dataset encompasses a total of 30 000 annotated objects, representing a significant scale that supports the validation of various algorithms and serves as a valuable resource for researchers both domestically and internationally.

B. Image Processing

During practical data acquisition, the images captured are often in color. However, color images contain a significant amount of information, which can increase the processing time. To mitigate this, it is common practice to convert color images to grayscale before conducting digital image processing. By doing so, the brightness distribution characteristics of the image can be preserved while reducing computational complexity and shortening processing time. In the case of surface defect detection, input images typically encompass both the defective target and the background. To effectively detect and analyze surface defects, it is crucial to separate the background from the target. This separation process allows for more accurate utilization and further analysis of the target, thereby facilitating the identification and comprehension of defects.

This study aims to address the issue of unwanted artifacts, such as conveyor belts, in the background of images captured by the camera. To achieve this goal, a Python-based image preprocessing pipeline is proposed in this article to transform the original image into one that only contains the paper without the interference of the conveyor belt background. This process involves multiple steps, including image reading, size adjustment, edge detection, contour finding, and image transformation, as shown in Fig. 15. The output image eliminates the interference caused by the conveyor belt background.

In these steps, the original image is first proportionally resized to a fixed height, followed by Canny edge detection. The Canny edge detection algorithm initiates its process by performing Gaussian blur to mitigate image noise. Subsequently, it employs Sobel filters to compute the gradient for each pixel, aiming to identify potential edge locations. Following this step, nonmaximum suppression is conducted, selectively retaining the most salient gradient values to refine the edges. The Canny algorithm then proceeds to employ high and low thresholds for discerning edge pixels, marking pixels with high gradient values as strong edges and those with values between the high and low thresholds as weak edges. Eventually, strong edges are connected to neighboring weak edges and retained. In the final phase, the Canny algorithm generates a binary image in which edge pixels are represented in white and nonedge pixels are represented in black all while providing comprehensive contour information.

Next, the method employs dilation and morphological operations to expand the regions in the binarized image for better identification of the smallest regions in the image. The four endpoints of all rectangular frames are obtained using the convex hull method on the largest contour points and returned. The rectangular corner points are then resorted in order, obtaining the new standardized corner positions of the rectangle and obtaining the coordinate array. Finally, a perspective transformation is administered to the initial rectangle, a procedure inherently grounded in matrix operations. Here, the coordinates of each pixel undergo a meticulous transformation contingent upon a specific set of transformation parameters. These parameters conventionally encompass elements of rotation, translation, scaling, and projection. Subsequently, the processed image is securely preserved onto the storage medium.

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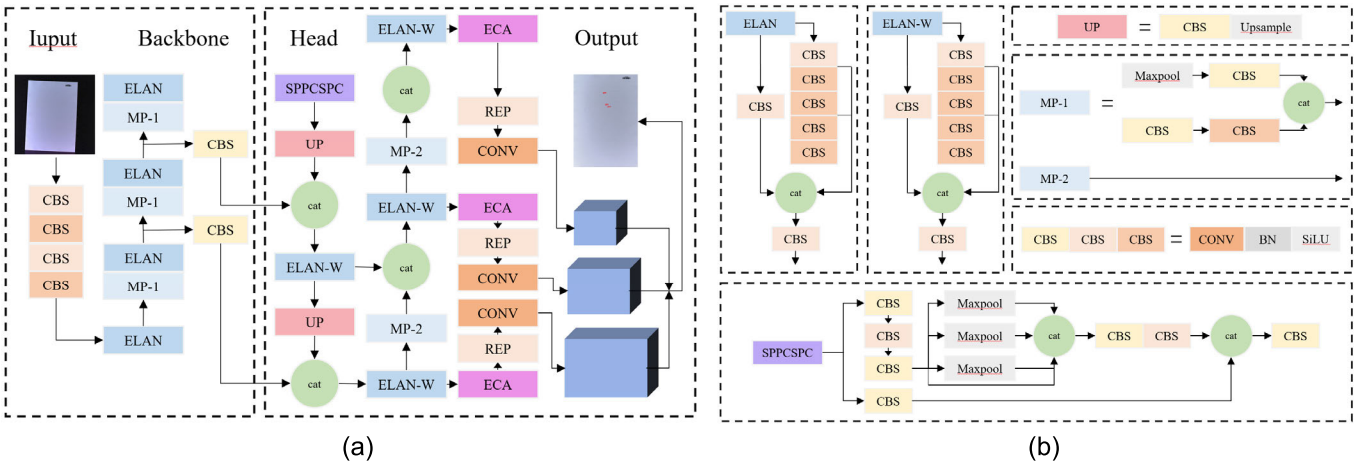


Fig. 13. Improved network structure diagram. (a) Architectural diagram of the improved YOLOv7 network. (b) Basic components.

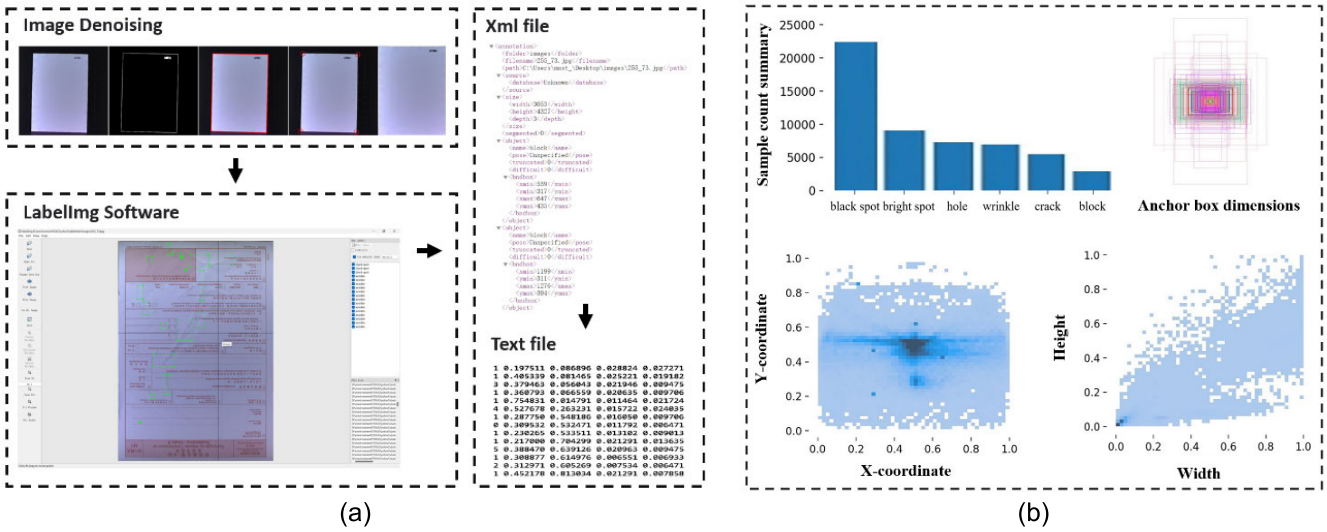


Fig. 14. Process of image labeling and information conversion. (a) Dataset creation process. (b) Dataset metrics.

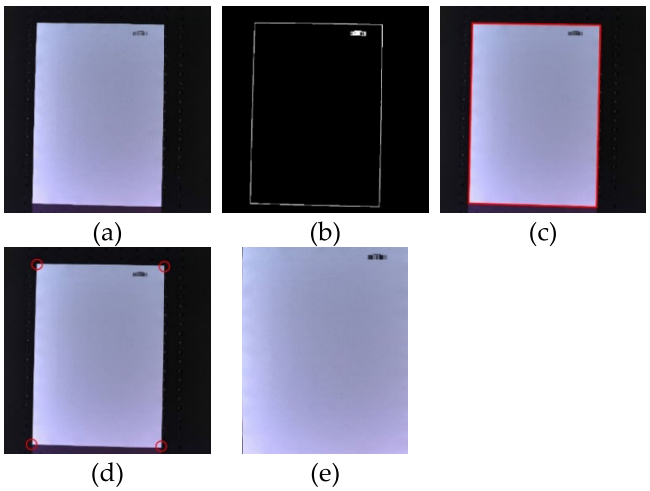


Fig. 15. Image processing. (a) Original. (b) Canny. (c) Max contour. (d) Box point. (e) Result.

This method exhibits a high degree of interference resistance. Background artifacts, such as conveyor belts, can greatly affect the accuracy and readability of the image. However, this method effectively eliminates such interference, thereby enhancing the accuracy and stability of image processing. Accurate image processing is crucial for ensuring product quality and improving production efficiency, particularly in the manufacturing industry. This method has broad application prospects and holds significant practical value in fields such as card recognition.

In addition to its application in card recognition, this method holds high practical value in various other fields. For example, in the pharmaceutical industry, this method can be utilized during the initial design phase of labels for drug bottles. It enables the transformation of the label images into ones that only contain the paper part, facilitating easy QR code scanning and recognition of the labels. Furthermore, this method can be applied to the detection of packaging boxes during the packaging process. By using this method, the captured images can be transformed into ones that only contain the packaging box, excluding other interfering elements. This reduction of unwanted elements simplifies the subsequent image processing tasks, reducing the difficulty and complexity involved.

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TABLE IV
SYSTEM HARDWARE AND SOFTWARE CONFIGURATION

Platform	Configuration
Central Processing Unit (CPU)	12th Gen Intel(R) Core(TM) i9-12900H 2.50 GHz
Graphics Processing Unit (GPU)	NVIDIA GeForce RTX 3070 Laptop GPU
Operating System	Windows 11
Integrated development environment (IDE)	PyCharm
Programming language	Python 3.7.15
Open Source Deep Learning Framework	PyTorch 1.11.0
Acceleration library	CUDA 11.3
GPU-accelerated library of primitives for deep neural networks	CuDNN 8.2.1

In conclusion, the Python-based image preprocessing pipeline is a promising image processing method that transforms the original image into one that only contains the regions of interest. It has broad application prospects. Furthermore, by continuously optimizing and improving the method, it can further enhance processing efficiency and accuracy, providing strong support for achieving intelligent manufacturing and digital transformation.

C. Configuration of the Experimental Environment

The experiments conducted in this article were carried out in a software environment based on the Windows 11 operating system. The deep learning environment utilized CUDA 11.3, the PyTorch 1.11.0 framework, and Python 3.7.15. In terms of hardware, an Intel[1] Core[2] i9-12900H CPU was employed, and model training was accelerated using the NVIDIA GeForce RTX 3070 Laptop GPU. For a comprehensive understanding of the model's configuration, the detailed parameters are presented in Table IV.

D. Evaluation Indicators

In the field of object detection, performance evaluation metrics primarily include FPR, FNR, AP, and mAP. In addition, in practical applications, detection speed FPS and model parameters are also taken into consideration [41].

Regarding data partitioning, it is divided into TP, FP, TN, and FN. These terms represent the correct identification of actual defects as defects, incorrect identification of nondefects as defects, correct identification of nondefects as nondefects, and incorrect identification of actual defects as nondefects. P represents the proportion of TPs among all samples identified as defects, while recall (R) represents the proportion of TPs among all actual defect samples. FPS represents the number of frames processed per second. The formulas for calculating these evaluation metrics are as follows:

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} \tag{11}$$

[1]Registered trademark.
[2]Trademarked.

TABLE V
TRAINING PARAMETERS

Parameter	Configuration
Training epochs	300
Batch size	8
Optimizer	SGD=1E-2, Adam=1E-3
Matrix Reasoning Ways	True

$$\mathrm{FNR} = \frac{\mathrm{FN}}{\mathrm{TP} + \mathrm{FN}} \tag{12}$$

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{13}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{14}$$

$$\mathrm{AP} = \int_0^1 P\,(R)\,dt \tag{15}$$

$$\mathrm{mAP} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{AP}_i \tag{16}$$

$$F1 = \frac{2 * P * R}{P + R} \tag{17}$$

$$\mathrm{FPS} = \frac{1000}{T_{\mathrm{preprocess}} + T_{\mathrm{inference}} + T_{\mathrm{NMS}}} \tag{18}$$

where N represents the total number of classes in the classification data, i represents the current data category, $T_{\mathrm{preprocess}}$ refers to the time required for preprocessing a single image, $T_{\mathrm{inference}}$ represents the inference time, and T_{NMS} denotes the postprocessing time.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

A. Network Parameter Configuration

The input image size was set to 640 ×640 pixels, and the number of training epochs was set to 300 during the experiment. To facilitate comparative analysis, a batch size of 8 was utilized, and the default parameter values of YOLOv7 were employed. To expedite the local convergence of the model, cosine annealing was employed to dynamically adjust the learning rate. The learning rate was initially set to a small value and gradually increased until reaching a predefined larger learning rate. This adjustment method aids in improving the convergence speed of the model. In this experiment, the learning rate was set to 1E−2, while the detailed parameters are presented in Table V.

B. Comprehensive Ablation Experiments

In order to further assess the impact of the improved algorithm on the detection performance of paper defect recognition, we conducted experiments on our custom-made dataset of paper defects using the same experimental setup and parameters. We introduced attention mechanisms such as SE, CBAM, ECA, and CA into the YOLOv7 model and conducted experiments on them [42]. The improved models, compared with the original YOLOv7 model, exhibited varying degrees of enhancement in terms of precision, recall, mAP, and $F1$ score. The performance comparison results are illustrated in Fig. 16.

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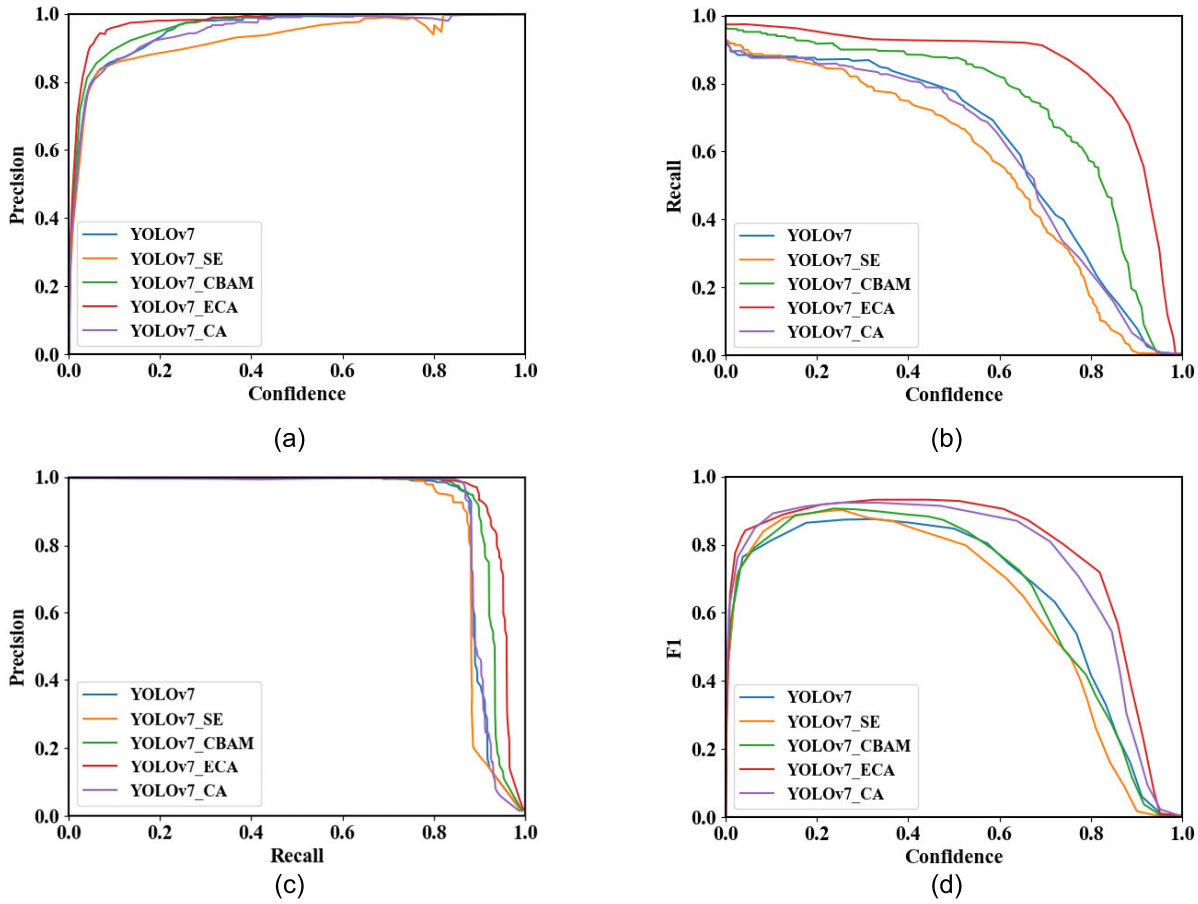


Fig. 16. Effect of attention mechanism on model. (a) Precision curve. (b) Recall curve. (c) Precision–recall curve. (d) $F1$-score curve.

Under the configuration set forth, all detection algorithms in this section can attain convergence.

To evaluate the accuracy of the dual-layer template, this study conducted detection on 10 000 images. Using the model equipped with the dual-layer template for paper detection, the visualization interface software displayed 7983 images as satisfactory and 2017 images as defective within this dataset [43]. Multiple manual inspections were performed, and performance metrics for the dual-layer template were calculated based on (13), (14), (16), and (17).

To further validate the optimization effects of the integrated modules in the improved YOLOv7 paper defect detection model based on the dual-layer template, we conducted ablation experiments [44]. The comprehensive results from the ablation experiments reveal that the incorporation of the dual-layer template, the FReLU activation function, and the ECA attention module has led to diverse degrees of enhancement in P, R, mAP, and $F1$, as elucidated in Table VI. In Table VI, the $\surd$ symbol is utilized to signify the addition of corresponding modules within the network model.

The results of the ablation experiments indicate that the components introduced in this study have led to varying degrees of improvement in the model's detection of targets. First, the dual-layer template method, which was added in the experiments, demonstrated the most significant overall enhancement, with approximately a 3% increase in P, R, mAP, and $F1$ on average. This implies that the dual-layer template-based approach effectively alleviates interference stemming from printed text in paper defects, consequently fostering a more expeditious and stable training process.

The impact of attention mechanisms on the network varies, and we selected the most effective ECA module. The FReLU activation function has a minor impact on the network but still offers an improvement of around 1%. Based on these collective experimental results and considering the requirements for accuracy and real-time performance in practical applications, the integration of the dual-layer template, FReLU activation function, and ECA attention module proves to be the most effective combination, resulting in substantial improvements in P, R, mAP, and $F1$, with increases of 7.7%, 4.5%, 7.9%, and 8%, respectively.

C. Comparative Experiment of Different Network Models

To validate the advantages of the proposed joint dual-layer template and attention mechanism in the paper product appearance detection model compared to other commonly used networks, we selected several popular models for performance comparison analysis and recorded the training results, as shown in Table VI. By observing the data in Table VII, we conclude that the proposed improved algorithm performs exceptionally well in terms of performance compared to mainstream single-step detection networks.

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TABLE VI
ABLATION EXPERIMENT RESULT

YOLOv7	Dual-Layer	FReLU	Attention Mechanism				Precision(%)	Recall(%)	mAP(%)	F1
			SE	CBAM	ECA	CA				
√	×	×	×	×	×	×	83.2	81.3	87.3	0.85
			√	×	×	×	81.2	79.6	83.6	0.87
			×	√	×	×	84.5	82.7	88.9	0.88
			×	×	√	×	86.3	83.3	89.6	0.88
			×	×	×	√	82.9	81.1	86.9	0.86
			√	√	×	×	83.1	80.9	85.6	0.87
			√	×	√	×	85.4	82.4	86.7	0.85
			√	×	×	√	82.8	80.2	84.3	0.84
			×	√	√	×	85.4	82.6	87.6	0.86
			×	√	×	√	83.6	81.6	86.5	0.87
			×	×	√	√	83.4	81.1	85.8	0.86
√	×	√	×	×	×	×	83.8	81.2	87.3	0.86
			√	×	×	×	81.1	79.9	84.6	0.84
			×	√	×	×	84.6	82.8	89.9	0.89
			×	×	√	×	86.7	83.6	90.3	0.88
			×	×	×	√	82.3	81.3	87.5	0.87
			√	√	×	×	83.6	81.2	86.6	0.87
			√	×	√	×	85.6	82.6	87.3	0.88
			√	×	×	√	82.9	80.4	84.2	0.86
			×	√	√	×	85.8	82.5	87.8	0.88
			×	√	×	√	83.9	81.5	87.2	0.88
			×	×	√	√	83.5	81.3	86.1	0.86
√	√	×	×	×	×	×	85.1	83.2	90.6	0.88
			√	×	×	×	83.5	81.5	86.7	0.87
			×	√	×	×	86.4	85.6	91.6	0.90
			×	×	√	×	89.9	84.6	94.5	0.93
			×	×	×	√	84.4	82.2	90.8	0.89
			√	√	×	×	85.6	80.7	89.7	0.88
			√	×	√	×	87.6	81.5	90.6	0.89
			√	×	×	√	85.2	83.3	87.9	0.87
			×	√	√	×	87.7	84.5	92.5	0.90
			×	√	×	√	85.6	83.7	91.4	0.91
			×	×	√	√	86.5	83.5	88.6	0.88
√	√	√	×	×	×	×	86.3	84.7	93.7	0.91
			√	×	×	×	84.1	83.6	90.9	0.89
			×	√	×	×	87.6	83.4	94.3	0.92
			×	×	√	×	**90.9**	**85.8**	**95.2**	**0.93**
			×	×	×	√	83.9	83.9	92.6	0.91
			√	√	×	×	86.6	82.7	91.8	0.89
			√	×	√	×	88.8	83.6	94.6	0.91
			√	×	×	√	86.3	84.3	93.9	0.91
			×	√	√	×	90.1	85.5	94.9	0.92
			×	√	×	√	85.6	84.9	92.8	0.91
			×	×	√	√	86.5	85.6	91.5	0.89

TABLE VII
COMPARISON OF EXPERIMENTAL RESULTS OF DIFFERENT NETWORK MODELS

Network model	Precision(%)	Recall(%)	mAP(%)	FPS
Faster-RCNN	84.2	72.3	89.1	25
RetinaNet	76.2	79.3	69.4	33
SSD	77.2	73.6	73.8	17
YOLOv3	79.2	69.1	79.6	32
YOLOv4	81.9	79.3	82.9	43
YOLOv5s	83.1	80.5	83.6	67
YOLOv7	83.2	81.3	87.3	82
Proposed Model	**90.9**	**85.8**	**95.2**	**85**

From the results of the ablation experiments, it is evident that, compared to two-stage detection algorithms such as Faster R-CNN, YOLO possesses rapid detection capabilities but is subject to certain limitations in terms of accuracy. The proposed algorithm in this article addresses these accuracy limitations and gains advantages in both detection accuracy and speed. It has improved P, R, mAP, and FPS by 6.7%, 13.5%, 6.1%, and 60, respectively. This algorithm effectively enhances the performance of paper defect detection and recognition while controlling parameter size.

D. Comparison With Models From Existing Literature

To validate the advantages of the proposed joint dual-layer template and attention mechanism in the paper product appearance detection model compared to other commonly used networks, we selected the results reported in [19] for performance comparison analysis and recorded the training results as shown in Table VIII. By observing the data in Table VII, we can conclude that the proposed improved algorithm performs exceptionally well in terms of performance compared to [19].

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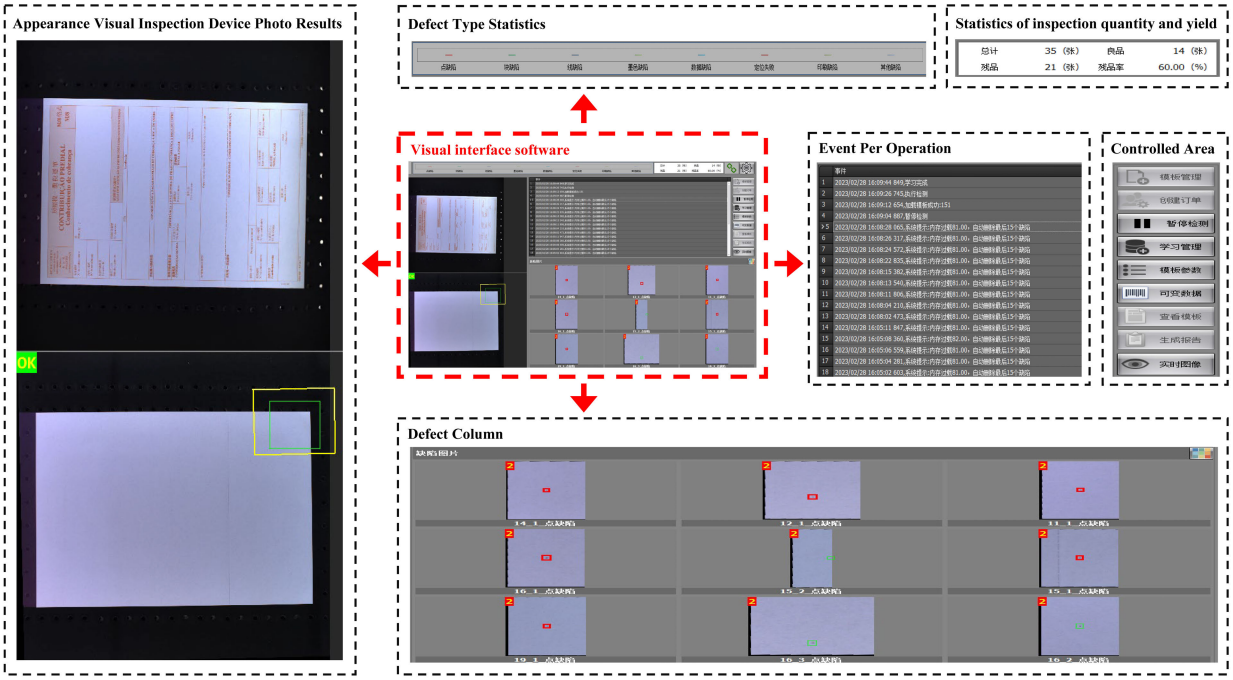


Fig. 17. Visual interface software verification.

TABLE VIII
COMPARISON WITH EXISTING MODELS

Network model	Precision(%)	mAP(%)
CNN + FMCC [19]	88.69	93.62
Proposed Model	**90.90**	**95.20**

TABLE IX
COMPREHENSIVE ANALYSIS OF PAPER SHEET DEFECT RECOGNITION PRECISION

Defect nomenclature	Quantity of defect images	Detecting defective images	Precision(%)
Black spot	1617	1568	96.97
Bright spot	88	87	98.86
Hole	3	3	100
Wrinkle	13	13	100
Crack	9	9	100
Block	36	36	100

E. Visual Interface Software Was Utilized to Validate the Detection Results

In order to facilitate a more convenient and intuitive way of assessing the condition of paper, this study developed a visual interface [45], as depicted in Fig. 17. The interface offers various options and functionalities. The left side of the interface displays the first and second cameras, which are used to detect the front and back sides of single-page paper products. In the top-right corner, the interface displays the number of satisfactory and defective paper products detected, as well as the calculated reject rate. The bottom-right corner provides real-time visualization of detected paper defects, enabling a precise understanding of their specific locations. This allows for continuous monitoring of progress and quality status, facilitating the prompt identification and resolution of issues.

The visual interface also enables automated control of the production process. The visual interface provides a real-time display of the quantities of satisfactory and defective products, as well as the reject rate. In addition to the mentioned functionalities, the visual interface offers more convenience and practicality [46]. Users can flexibly set parameters such as paper size, quantity, thickness, and quality to better adapt to different types of paper products and ensure the accuracy of the detection results.

This flexibility allows the system to meet various paper production requirements and provide users with customized detection solutions. As an example, the interface can be configured to automatically stop operations once the desired quantity of satisfactory paper, say 100 000, is reached, streamlining the operational workflow and improving efficiency, as illustrated in Fig. 18. After testing on over 100 000 newly printed paper sheets, defect quantity statistics reveal that, out of 101 523 sheets inspected, 100 000 were deemed to be of satisfactory quality, while there were 1523 substandard sheets. Testing results on the test set exhibit variations in the recognition of six paper defects, as depicted in Table IX. High-resolution defects, such as hole, wrinkle, crack, and block, which occupy a larger proportion of the target pixels, achieved a recognition accuracy of 100%. The recognition accuracy for black spots was 96.97%, and for bright spots, it was 98.86%.

An analysis of the detection data indicates that the recognition accuracy for black spots is relatively lower, primarily due to the extremely low pixel proportion of black spot targets within the paper, making them more challenging to detect. The test results indicate that the model's accuracy on the test set

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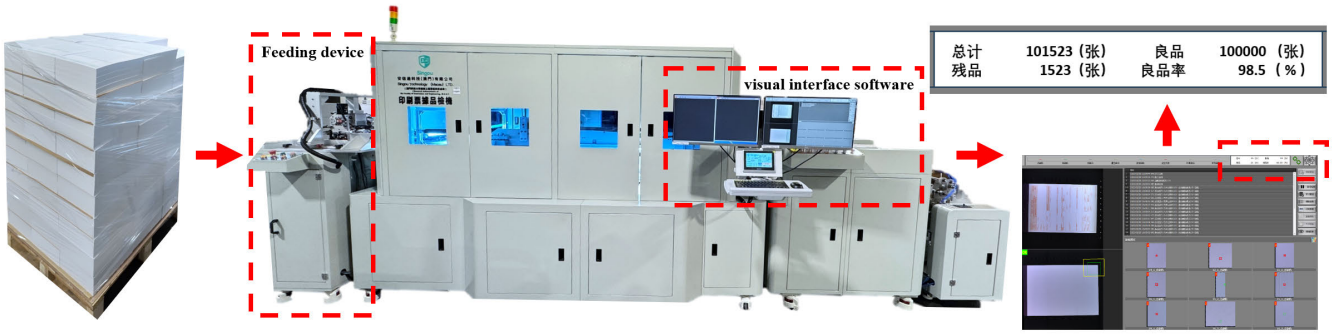


Fig. 18. Completing the inspection process for the collection of 100 000 good products.

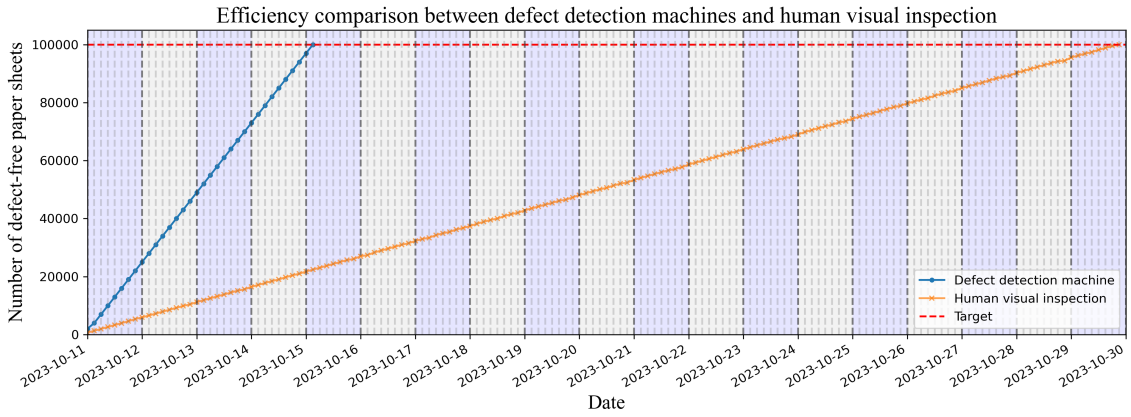


Fig. 19. Efficiency comparison between defect detection machines and human visual inspection.

significantly surpasses the results on the training and validation sets, which could be attributed to the accumulation of a greater quantity of training image data in practical applications.

The visual interface not only provides real-time monitoring and recording of data during the production process but also offers data statistics and analysis capabilities, including information on paper detection results and equipment status. By enabling real-time monitoring and data recording, the visual interface assists factory managers in promptly responding to issues, thereby improving production speed and adaptability. It conducts data analysis specific to different paper products, helping users gain deeper insights into the production process and identify problems and bottlenecks. Through data analysis, users can make timely adjustments to production parameters and workflows, enhancing efficiency and reducing costs.

This study conducted a comprehensive experimental investigation into the comparative efficiency of defect detection machines and human visual inspection. The experiments were carried out at the Printing Bureau of Macao Special Administrative Region of the People's Republic of China, ensuring the credibility and practical applicability of the findings. Both groups were tasked with collecting 100 000 flawless paper sheets, commencing on October 11, 2023. Work hours were reasonably scheduled, spanning 8 h per day from 9:00 A.M. to 1:00 P.M. and from 2:00 P.M. to 6:00 P.M. Moreover, data collection during the experiments was meticulous, as illustrated in Fig. 19. The quantity of defect-free paper sheets collected was recorded hourly, facilitating an accurate assessment of performance variations between the two groups.

The result data clearly demonstrate that defect detection machines exhibit higher efficiency and stability. The number of defect-free paper sheets completed by the machine per hour shows a more stable and consistent trend with minimal fluctuations. This indicates that the machine is hardly affected by subjective factors during the detection task, rendering it highly reliable. What is most impressive is that the defect detection machine accomplished the collection of 100 000 defect-free paper sheets in approximately four working days, while human visual inspection required a full 19 working days to achieve the same quantity of inspection. This substantial difference in time efficiency vividly highlights the efficiency and practicality of defect detection machines.

The visual interface supports two modes of operation: online detection and offline detection. In the former, it performs a real-time detection of paper and provides immediate feedback on the results, facilitating timely adjustments to the production process. The offline detection mode, on the other hand, conducts paper detection when the production line is paused, and the results can be processed once production resumes, thereby improving production efficiency and saving time. The visual interface serves as not just a control platform for the visual inspection device but also the core of intelligent management for the entire production line. It plays a crucial role throughout the production process, allowing for easy monitoring of paper status. Through automation control, remote monitoring, and data analysis, the interface enhances production efficiency and quality while reducing the risk of human errors and lowering production costs. It drives the digitalization and intelligent development of the production

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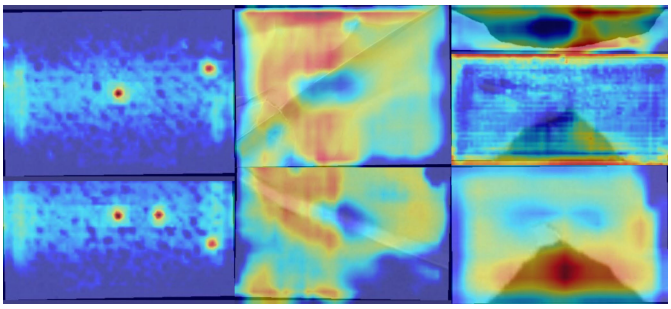


Fig. 20. Yolov7 heat map before improvement.

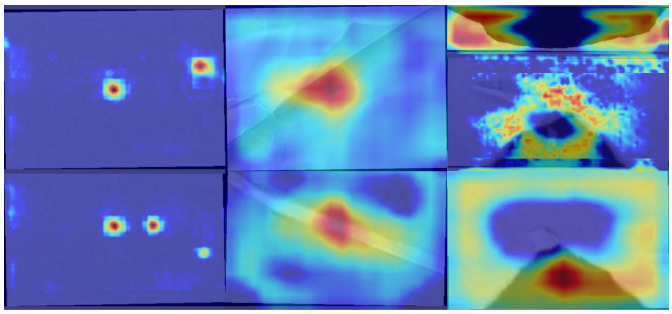


Fig. 21. Yolov7 heat map after improvement.

line, making significant contributions to the transformation and upgrading of the manufacturing industry and the advancement of smart manufacturing.

F. Heat Map Verification

In order to intuitively evaluate the improved algorithm of the paper defect detection system proposed in this article, a visual interface is developed, which uses a heat map to describe the density, distribution, and changes in data groups on the page. A heat map is a technique in computer vision that visualizes the regions of interest in image classification and object localization tasks performed by deep learning models [47].

By generating a heat map, we can intuitively observe the model's focus on different regions. Heat maps are widely used in computer vision for visualizing prediction results, determining location information, explaining the model's decision-making process, and predicting outcomes. In addition, heat maps provide valuable insights for image analysis and understanding tasks. Common methods for generating heat maps include CAM and Grad-CAM. CAM utilizes global average pooling to learn feature weights, reconstruct target positions, and generate corresponding heat maps. Grad-CAM improves upon CAM by incorporating gradient information to assign weights to each pixel, resulting in more accurate localization of target regions. This makes Grad-CAM suitable for complex network architectures.

As observed in Fig. 20, it is evident that the heatmap of the original model displays relatively scattered highlighted regions, indicating an excessive and unnecessary focus outside the target areas. However, following the introduction of the dual-layer template and attention mechanism, significant improvements are observed in Fig. 21. The new heatmap exhibits more concentrated highlighted regions, primarily centered around the positions of the target objects. This enhancement enables the network to comprehensively capture essential features, enhancing its sensitivity to the location information of small objects for more precise localization.

Furthermore, the network's attention focus becomes more centralized, facilitating a comprehensive consideration of the entire area of the target object and improving its ability to focus on defect targets. This improvement effectively suppresses interference from background features, further enhancing the detection performance for paper defect targets. It substantiates the effectiveness of the proposed improvements.

VII. Discussion

Although the paper defect detection system in this article demonstrates superiority in comparative studies and ablation studies, there are still some limitations. Due to the extremely small size of paper defects, high demands are placed on the camera. In the future, paper scanning can be considered as an approach to improve data accuracy. The future research directions of the paper defect detection system include the following aspects: first, exploring more advanced algorithms and models to enhance the accuracy and robustness of the system; second, conducting multimodal data fusion using various sensors or data sources to enhance the detection capability of the system, and by combining different types of data, a more comprehensive understanding and characterization of paper defects can be achieved; and furthermore, building a more comprehensive dataset of paper defects and developing comprehensive evaluation criteria to compare and evaluate the performance of algorithms. This will drive the development of paper defect detection technology to meet the ever-changing demands of paper production.

VIII. Conclusion

A novel system, combining dual-layer templates and attention mechanisms, is proposed for the visual inspection of paper products, aiming to effectively address surface defects encountered in print production. The system's development includes the creation of a visual interface that provides a customized defect detection solution for the printing authority. This solution possesses the capability to automatically detect and eliminate flawed paper products, thereby elevating work efficiency and reducing labor costs. To overcome the limited field of view of array cameras when dealing with larger paper sizes, a synergistic approach incorporating controllers and paper detection sensors has been introduced. By matching template images with the images to be inspected and leveraging an enhanced YOLO algorithm, the system notably improves both the accuracy and speed of detection. Comprehensive ablation experimental results reveal an average precision of 95.2%, representing a significant 7.9% enhancement over the original algorithm. This system achieves the inspection of paper product appearances while also offering automation control, remote monitoring, and data analysis functionalities, thus propelling the transformation and advancement of intelligent manufacturing within the manufacturing industry.

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Xuanfeng Li received the B.S. degree in automation from the Southeast University Chengxian College, Nanjing, China, and the B.S. degree in human resource management from Nanjing University, Nanjing, in 2022. He is currently pursuing the M.S. degree in intelligent technology with the Macau University of Science and Technology, Macau, China.

His research interests include computer vision and deep learning.



Haibo Yan received the B.Sc. degree in aircraft manufacturing engineering from Shenyang Aerospace University, Shenyang, China, in 2000, and the M.Sc. degree in industrial engineering from Shanghai Jiao Tong University, Shanghai, China, in 2015. He is currently pursuing the Ph.D. degree with the Macau Institute of Systems Engineering, Macau University of Science and Technology, Macau, China.

His current research interests include applied mathematics, visual measurement, and visual positioning.



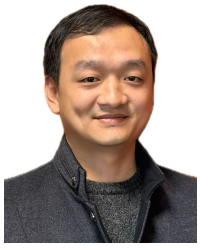
Kai Cui was born in Shanxi, China, in 1995. He received the B.S. degree in electronic science and technology and the M.S. degree from the North University of China, Taiyuan, China, in 2018 and 2021, respectively. He is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering, Macau University of Science and Technology, Macau, China.

His research interests include deep learning, object detection, time series forecasting, and computer vision.



Zhiwu Li (Fellow, IEEE) received the B.S. degree in mechanical engineering, the M.S. degree in automatic control, and the Ph.D. degree in manufacturing engineering from Xidian University, Xi'an, China, in 1989, 1992, and 1995, respectively.

In 1992, he joined Xidian University. He was a Visiting Professor with the University of Toronto, Toronto, ON, Canada. He was with the Technion—Israel Institute of Technology, Haifa, Israel; the Martin-Luther University of Halle–Wittenburg, Halle, Germany; Conservatoire National des Arts et Métiers, Paris, France; and Meliksah Universitesi, Kayseri, Turkey. He is currently with the Institute of Systems Engineering, Macau University of Science and Technology, Macau, China. His current research interests include Petri net theory and application, supervisory control of discrete-event systems, workflow modeling and analysis, system reconfiguration, game theory, and data and process mining.



Ruibin Liu received the M.B.A. degree from the Macau University of Science and Technology, Macau, China, in 2019, where he is currently pursuing the Ph.D. degree with the Institute of System Engineering.

His current research interests include machine learning and system dynamics model.



Guibin Lu is currently pursuing the Ph.D. degree with the Macau Institute of Systems Engineering, Macau University of Science and Technology, Macau, China.

His research interests include deep learning and big data analysis.



Kai Chin Hsieh received the B.Eng. degree in mechanical engineering with business finance from University College London, London, U.K., in 2023, where he is currently pursuing the M.Eng. degree.

He is particularly captivated by the potential of computer vision in sports analytics, with a special focus on precision tracking and motion analysis. Given the interplay of engineering and finance in his academic background, he is also keen on exploring how technological innovations can drive efficiencies in business models. In addition, he is enthusiastic about harnessing the power of machine learning in enhancing real-time data acquisition and interpretation, aiming to bridge the gap between engineering solutions and their practical financial implications. His research interests intersect the domains of mechanical engineering and advanced computational technologies.



Xiaoshi Liu is currently pursuing the B.Sc. degree in business administration with the Macau University of Science and Technology, Macau, China.

His research interests include statistical management.



Chitin Hon (Senior Member, IEEE) received the B.S. degree in electronics from Guangzhou University, Guangzhou, China, in 1996, the M.S. degree in computer software application from Jinan University, Guangzhou, in 2000, and the Ph.D. degree in management science and engineering from the South China University of Technology, Guangzhou, in 2008.

Since 2003, he has been with the Macau University of Science and Technology, Macau, China, where he is currently an Associate Professor with the Faculty of Innovative Engineering. His current research interests include pandemic and epidemic dynamics and prediction, human–robot interaction, and statistical learning.

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